FAST-National University of Computer and Emerging Sciences (NUCES)

Literature Review

Project Topic:

Small Language Models (SLMs) in Resource-Constrained Multi-Agent Systems

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Executive Summary

The rapid expansion of Large Language Models (LLMs) has revealed critical bottlenecks in API expenditures, operational latency, and data privacy. Consequently, research interest has shifted toward deploying Small Language Models (SLMs) (1B–7B parameters) running locally on commodity workstations and personal computers. To compensate for the reduced cognitive capacity of individual SLMs, researchers investigate collaborative Multi-Agent Systems (MAS). This literature review synthesizes research from 2023 to 2026 across model routing, multi-agent debate topologies, and local privacy-preserving workflows. By evaluating the methodologies, datasets, and limitations of eight key studies, we identify operational gaps in sensitivity-aware routing and localized agent communication overhead, framing the direction for a simulated local multi-agent privacy framework.

1 Introduction

Frontier Large Language Models (LLMs) achieve strong benchmark reasoning but introduce significant operational overhead. Cloud API dependencies entail recurring financial costs and expose sensitive user prompts to external servers. Conversely, Small Language Models (SLMs) can run locally on consumer-grade hardware, such as standard laptops and university lab workstations with limited RAM and VRAM.

Although SLMs drastically minimize inference cost and preserve data sovereignty, their isolated performance on multi-hop reasoning and long-horizon planning remains limited. Multi-Agent Systems (MAS) address these deficits by distributing complex workflows across specialized, interacting model instances. This review surveys and synthesizes foundational literature (2023–2026) regarding hybrid routing, collaborative multi-agent reasoning, and client-side privacy architectures, establishing the empirical grounding for a simulated multi-agent system evaluated on standard PC hardware.

2 Synthesis of Existing Literature

Rather than summarizing each publication in isolation, this review categorizes and critically contrasts the literature across three interdependent domains: Dynamic Model Routing, Collaborative Agent Debate, and Localized Privacy Architectures.

2.1 Dynamic Model Routing and Cascade Systems

A primary focus in resource-constrained systems is reducing dependency on expensive models by routing simpler queries to smaller architectures.

Chen *et al.* [1] introduced **FrugalGPT**, addressing escalating inference costs via sequential cascade routing. In their framework, user prompts are sequentially dispatched to progressively larger models, halting as soon as an internal generation confidence metric surpasses a set threshold. Evaluated across datasets including GSM8K and HEADQA, FrugalGPT cut API costs by up to 98% while matching individual frontier model accuracy. However, its primary limitation lies in its heuristic confidence scoring: in open-ended or ambiguous tasks, models often exhibit miscalibrated confidence, causing early termination on hallucinated answers.

Ong *et al.* [2] advanced this concept with **RouteLLM**, shifting from post-hoc cascade thresholds to predictive, pre-execution routing. Using preference judgment datasets from the LMSYS Chatbot Arena, they trained matrix factorization and embedding-based classifiers to predict whether a query required a frontier model or could be solved by an open-weight SLM (e.g., Mixtral). On MT-Bench and MMLU, RouteLLM reduced cloud LLM calls by 86% while preserving 95% of GPT-4 benchmark performance. Nonetheless, RouteLLM operates on static single-turn inputs; it lacks state-tracking mechanisms to route queries within multi-turn, multi-agent conversational contexts.

2.2 Multi-Agent Collaboration, Debate, and Pruning

To elevate the reasoning ceiling of smaller models without increasing parameter scale, researchers leverage structured interaction topologies.

Du *et al.* [3] investigated multi-agent debate to mitigate hallucinations and factual drift. In their architecture, multiple distinct model instances generate candidate solutions and iteratively critique their peers' reasoning across multiple rounds. Evaluated on MMLU and grade-school math benchmarks, consensus-driven debate substantially improved factual accuracy over single-agent chains-of-thought. However, the study relied on uniform, high-parameter models hosted via cloud APIs; the authors did not explore the execution of heterogeneous SLMs running within the shared memory constraints of a single physical machine.

Liu *et al.* [4] introduced **DyLAN** (Dynamic LLM-Agent Network) to tackle the communication bloat of static agent frameworks. By conceptualizing the agent ensemble as a dynamic feed-forward computational graph, DyLAN scores agent contributions at runtime and prunes redundant nodes. Benchmarked on MATH and HumanEval, DyLAN outperformed static peer networks while improving token efficiency. Its main limitation is that the node selection scoring relies on complex ranking passes, which introduce compute bottlenecks when executed on local CPU/GPU runtimes.

Li *et al.* [5] analyzed scaling properties in "More Agents Is All You Need," proving that simple sampling-and-voting ensembles of smaller models can match the output quality of significantly larger counterparts on reasoning tasks like GSM8K. However, their

findings assume unbounded execution concurrency. On standard local hardware, parallel instantiation of multiple agent instances causes context-switching latency and memory fragmentation.

2.3 Local Privacy and Agentic Orchestration

Deploying language systems within privacy-sensitive domains (e.g., enterprise documentation, academic records, and healthcare) demands local containment.

Gorenshtein [6] presented a comprehensive systematic review on utilizing SLMs ($\leq$4B parameters) for clinical information extraction, evaluating 14 studies using datasets such as MIMIC-III. The synthesis showed that domain-adapted SLMs executed on local commodity GPUs achieve up to 91% parity with frontier models for structured entity extraction. The critical limitation identified is the fragility of isolated SLMs: without secondary agent verification or guardrails, error propagation in unsupervised local tasks remains unacceptably high.

Xi *et al.* [7] and Wang *et al.* [8] formulated architectural surveys detailing planning, memory, and tool utilization across autonomous agents. Both studies observed that while large models perform autonomous planning reliably, smaller models require constrained, deterministic communication frameworks (e.g., state graphs) to avoid diverging during complex multi-step routines.

3 Comparison of Selected Papers

4 Identified Research Gaps

The synthesis of existing work reveals two clear academic gaps:

1. **Absence of Privacy-Aware Semantic Routing:** Existing frameworks such as FrugalGPT [1] and RouteLLM [2] optimize model selection strictly along axes of *task difficulty* and *cost*. They do not assess *data confidentiality*. Consequently, current pipelines either expose sensitive prompt data to cloud APIs or restrict the entire workflow to local hardware, sacrificing advanced reasoning capacity.

2. **Lack of Verified SLM Swarm Architectures on Commodity Hardware:** While multi-agent debate [3] and ensembling [5] establish theoretical reasoning improvements, existing benchmarks assume unbounded compute clusters or cloud APIs. The literature lacks empirical benchmarks evaluating multi-agent peer verification where multiple quantized SLMs execute sequentially on standard local computer resources (e.g., 16GB system RAM, CPU-only or consumer GPU).

5 Proposed Project Direction

To address these limitations, this project proposes a **Local Privacy-Preserving Multi-Agent Framework** simulated and benchmarked entirely on commodity PC hardware.

The proposed system utilizes a local, quantized SLM (e.g., Llama-3.2-1B or Qwen-2.5-1.5B running locally via an in-memory runtime) operating as a Semantic Redaction and Routing Agent. The local agent scans incoming user prompts, strips Personally Identifiable Information (PII) using structured contextual replacements, and assesses task difficulty. If high-order analytical reasoning is needed, the sanitized payload is dispatched to a frontier cloud model. Simpler tasks or sensitive summaries are retained locally and processed by an internal SLM peer-review loop to suppress hallucinations without external data exposure.

We will simulate and measure this pipeline against monolithic cloud baselines, tracking PII leak rates, factual retention scores, and end-to-end execution latency on standard PC hardware.

References

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Table 1: Comparative Analysis of Literature (2023–2026)

Paper	**Core Problem**	**Method / Architecture**	**Dataset(s)**	**Limitations**
Chen [1]	API Cost Overhead	Sequential cascade routing based on output confidence.	GSM8K, HEADQA	Unreliable confidence calibration in open-ended tasks.
Ong [2]	Query Model Selection	Pre-execution routing via matrix factorization classifiers.	MT-Bench, MMLU	Limited to single-turn requests; ignores agent state.
Du [3]	Hallucinations	Multi-round agent debate with cross-examination.	MMLU, Math	Evaluated using large cloud models; high token cost.
Liu [4]	Agent Redundancy	DyLAN: Dynamic agent selection and graph pruning.	MATH, HumanEval	High computational overhead for runtime agent scoring.
Li [5]	Limits of SLMs	Ensembling and majority voting over small models.	GSM8K, HumanEval	High local memory consumption during concurrency.
Gorenshtein [6]	Sensitive Data	Localized sub-4B SLM processing on single GPUs.	MIMIC-III	Lacks peer-review agents; isolated SLMs vulnerable.
Xi [7]	Structural Fragility	Unified taxonomy of planning, profiling, and memory.	Survey	Lacks structured coordination protocols for SLMs.
Wang [8]	Coordination Limits	Systematic survey on cognitive architec-tures/workflows.	Survey	Notes absence of sensitivity-aware dispatching.